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Sequences reproduce cortical bidirectional connectivity statistics

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Abstract

In this work, we focus on the statistics of connectivity in a neural network optimised for storing sequences. Experimental cortical connectivity is shown to be well reproduced by a neural network model optimised to store patterns as fixed points (attractors), while the experimental values cannot be reproduced by a network optimised to store a sequence of uncorrelated patterns. We first look at a sequence of correlated patterns being optimally stored in the Hopfield network model. Further, we look at a recurrent neural network with weights learnt using the perceptron learning rule to store a sequence of correlated patterns. We explore the connectivity statistics of both the models and find them to be unable to reproduce cortical connectivityy. We then explore a different paradigm of information storage, namely a set of correlated sequences, using the same methods and find that in a network that optimally stores a set of correlated sequences, the experimental cortical connectivity is well reproduced.

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1 Introduction

Recurrently connected networks (RNNs) may serve as a potential model for the neocortex, as studies have shown that local networks of pyramidal cells in the neocortex are extensively interconnected through recurrent connections. Investigating the nature of the information stored in such networks is an ongoing debate. A popular way of looking at the problem of information storage is in terms of attractor states, while another way of looking at it is in terms of patterns stored as a sequence. An archetypal model of the attractor state network is the Hopfield network. A Hopfield network is a fully connected recurrent neural network with a simple Hebbian like setting of weights, popular due to its simplicity and ability to store patterns, upto a certain capacity. The Hopfield network can also be modified to store a sequence of patterns, and at the resulting capacity, the weight distribution can be studied to see whether it can be used as a good model for reproducing the recurrent connectivity of the neocortical network.

For a different approach, one can look at recurrently connected networks with zero welf-weights and the rest of the weights learned by the perceptron learning rule.[1] The perceptron learning rule is shown to be optimal and guaranteed to be able to learn the output from the input patterns if such a solution exists. The analogy between this and the Hopfield network is that we can think of each single neuron in the Hopfield network to be the output neuron of a perceptron and every other neuron being the input layer, where the input patterns are presented. There are no hidden layers, this is a single layer perceptron. We effectively thus have N perceptrons, where N is the number of neurons in the network. Modifications to the perceptron learning rule can lead to models of different types of networks (positive excitatory weights and fixed inhibitions is what we concentrate on here). In this project, we focus on a neural network with weights adjusted by a modified perceptron learning rule to look at different paradigms of information storage. We are comparing attractors, a sequence of correlated patterns and finally a set of sequences of correlated patterns. This model of the cortical network that has been shown to faithfully reproduce cortical bidirectionality statistics when it is optimised to store patterns as attractors. We investigate models which are optimised to store other kinds of information.

2 The Hopfield network for a sequence of patterns

The Hopfield network for a sequence of patterns consists of a network of N neurons, where each neuron can have only two states, which are taken to be active and inactive. We can choose to encode the states as $1/0$ or as $1/-1$ depending on whether we want to think of 'spike' like variables σ or of states S, respectively, but for the following dynamics, we can always map to $1/-1$ for simplicity. Neurons interact with each other by the weights w_{ij} connecting every neuron to the other. There are no self weights in the network. The input potential of a neuron i influenced by other neurons in the network is given by $h_i(t) = \sum_j w_{ij} S_j(t)$ and the dynamics of the network evolve in discrete time steps Δt, according to the following equation, where we have taken deterministic synchronous dynamics.

$$S_i(t + \Delta t) = sgn[h(t)] \tag{2.1}$$

The patterns being presented consist of $\xi_i^\mu \in 0, 1$ and the weights of the network are taken to be according to the following Hebbian like setting, for storing a sequence

$$w_{ij} = c' \sum_{\mu=1}^{M} (\xi_i^{\mu+1} - b)(\xi_j^\mu - a) \tag{2.2}$$

where a is the average sparsity or coding level of each pattern, which has been taken to be 0.5 for all our simulations. The value of b can be taken to be equal to a or can be taken to be 0, or we can try to optimise the value as a learnt parameter based on how well we can recall a sequence. The value of c is taken as $\frac{1}{2a(1-a)N}$. The overlap of a pattern in the sequence with the state of the Hopfield network during the recall is defined as $m^\mu = \frac{1}{2a(1-a)N} \sum_j (\xi_j^\mu - a) S_j$. We let our Hopfield network reach its optimal condition of storing a sequence of correlated patterns, by deciding the maximal capacity to be when on being initiated by the first pattern of the sequence, the network exactly recalls, at each step of the dynamics, the corresponding pattern in the sequence. We call a pattern being recalled if the overlap of the state of the Hopfield network with that pattern is above a certain high threshold that allows for some neurons to be wrong but most of them to be in the correct position.

2.1 Methods

A sequence is generated using a Markovian process as detailed by Clopath et al.[2]. Given a certain pattern of sparsity 0.5, the next pattern in the sequence, having a correlation $c \neq 0$

is produced as follows : The probability of an element in the next pattern changing its value compared to the current pattern is c while the probability of it being the same is $1 - c$. For zero correlation, we have random patterns being stored in the sequence. We impose a certain tolerance limit for the actual Pearson correlation between successive patterns in this Markovian sequence, so that in the event of high correlations, we do not have the same pattern being incorporated into the sequence. These patterns are then used to set the weights. The capacity is calculated, as mentioned above, the maximum length of the sequence when the entire pattern is recalled. As mentioned in [3], we can separate the weight matrix into excitatory and inhibitory parts and the weight matrix we look at is only the excitatory portion of the weight matrix. At capacity, we binarize the excitatory weight matrix as $w^b_{ij} = \Theta(w_{ij} - w_c)$ where w_c is taken to 0.1 times the average weight. Then we calculate the connection probability and the bidrectional connection probability.

2.2 Results

The capacity calculation shows that the capacity drastically decreases as the correlations between patterns increases. This might be intuitively due to the reason that the dynamics of the Hopfield nework does not allow it to differentiate between very highly similar patterns during the evolution of the states. The capacity at higher network sizes is shown to be almost the same regardless of the size. This is shown in the figure 2.1 At the maximum capacity, we calculate

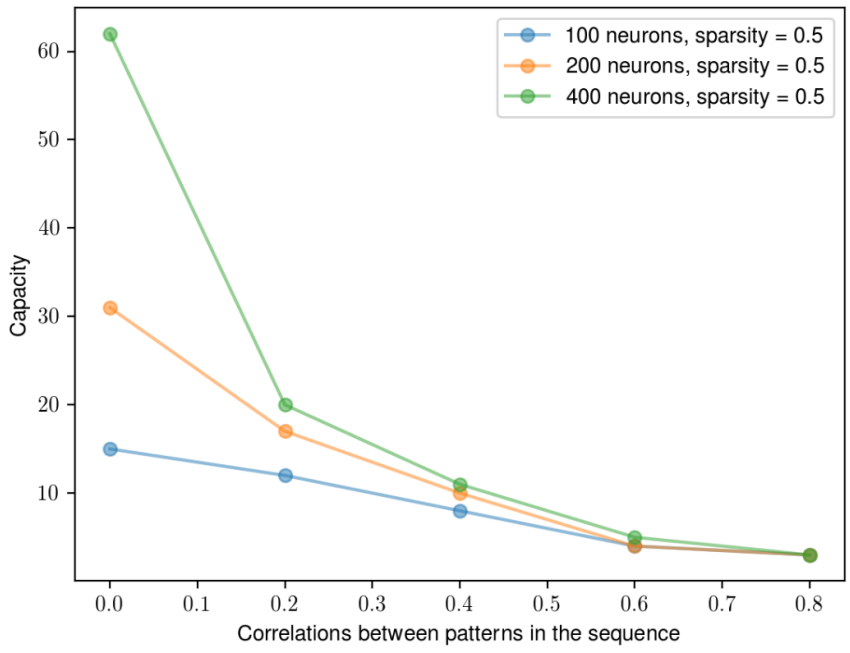

FIGURE 2.1: The capacity of the Hopfield network storing sequences, at increasing correlations

the bidirectional connectional probability versus the connection probability and find out that the weights eventually get more symmetric and approach the case of perfect symmetricity of the Hopfield network storing attractors, as shown in Figure 2.2

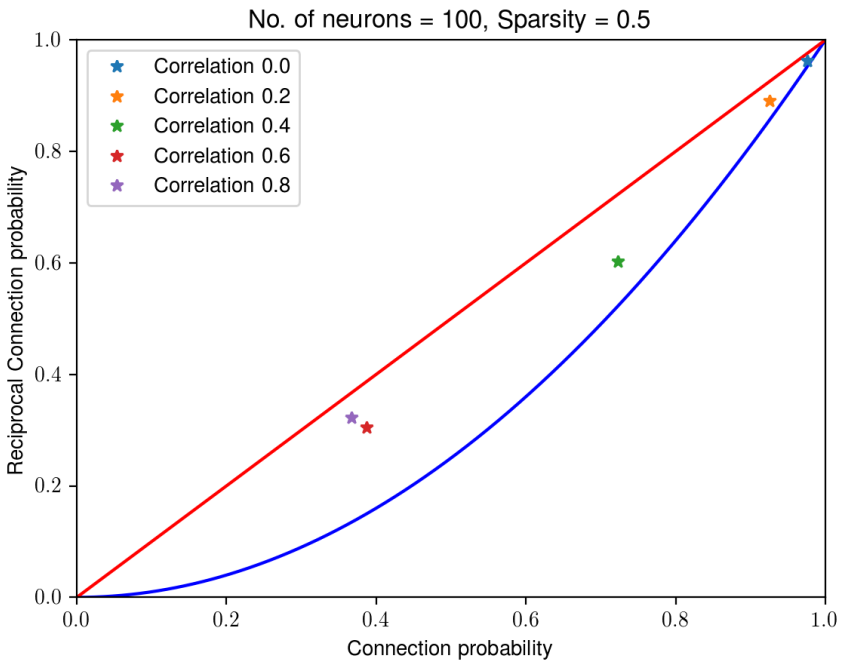

FIGURE 2.2: At maximum capacity, as we increase correlation, the bidirectional connection probability reaches the case of perfect symmtery

3 A neural network for sequence learning

The next thing we looked at are Hopfield like recurrently connected neural networks with the weights being learnt by a perceptron like learning rule. This was the crux of the project as we have a benchmark for comparing the results of this situation vis vis the simulations mentioned in Brunel[1]. Brunel calculated analytically, using both the cavity method and the replica method, the distribution of weights as well as the statistics of the weight matrix for attractor networks with uncorrelated patterns as well as for networks storing a sequence of uncorrelated patterns. He showed, both analytically as well as with simulations, that in the case of attractors, there are some characteristic features of the weight distribution as well as the weight matrix which are similar to the synaptic weight distribution obtained from experimental data. By tweaking the parameters in the model, (namely the coding level or sparsity and the attractor basin size) he has attempted to get close to some of the experimental data which might lead to the answer of the question whether neocortical connectivity is optimal for storing information in terms of patterns in an attractor network scenario or in terms of sequences.

However, information in terms of uncorrelated sequences may not be the best candidate to rival the popular paradigm of information being stored in terms of patterns in an attractor network. The second order statistics of the weight matrix for storing uncorrelated sequences is shown to be quite different from that of attractors. While in the attractors, there is significant overrepresentation of bidirectionally connected neurons, this is not so in the case of storing uncorrelated sequences. In terms of the simulations, this lack of overrepresentation of bidirectionally connected pairs of neurons can easily be seen in the figure. An intuitive argument applied by Brunel goes as follows : the two synaptic weights, w_{ij} and w_{ji} are driven by independent sources. While for the transition from pattern μ to pattern $\mu + 1$, the connection from neuron i to neuron j has to associate ξ_i^μ with $\xi_j^{\mu+1}$, the connection from neuron j to neuron i has to associate ξ_j^μ to $\xi_i^{\mu+1}$. Either of these associations is uncorrelated with the values of ξ seen by the connection in the opposite direction. This is in contrast to the case of attractors, for which we have, for one pattern in the attractor, each of the two reciprocal associations being perfectly correlated with the values of ξ as seen by the opposite direction. It is between these two extremes that we might place our paradigm - that of the storage of correlated sequences - because we notice that, intuitively, in the case of temporally correlated sequences, there is, in fact a correlation between the values of ξ as seen by the reciprocal connections, because for the transition from pattern μ to pattern $\mu + 1$, the connection from neuron i to neuron j has to associate ξ_i^μ with $\xi_j^{\mu+1}$, while for the transition from pattern $\mu + 1$ to pattern $\mu + 2$ the connection from neuron j to neuron i has to associate $\xi_j^{\mu+1}$ to $\xi_i^{\mu+2}$, and the two patterns

ξ_i^μ and $\xi_i^{\mu+2}$ can be correlated (in fact it is so in our paradigm). Our hypothesis is that with increased correlations between the patterns in the sequence we might get connectivity matrices which would have statistics that are closer to the optimal in terms of information storage.

3.1 Methods

As mention in [1], cortical networks are composed of excitatory and inhibitory neurons. Some evidence suggests that the excitatory subnetwork is highly structured, while connections involving inhibitory interneurons are more stereotyped, with close to full local connectivity. We therefore focused on the excitatory connectivity and incorporated inhibition in the thresholds of excitatory neurons. For a network learning sequences, we consider sequences of random patterns with a threshold T and a basin size K (we will take it to be mostly zero in the case of our simulations and take it at higher values when we wish to replicate more biologically plausible connection probabilities), and a coding factor or sparsity of 0.5. They are generated in the same manner as described in 2.1. The network should learn to retrieve robustly the whole sequence, starting from the first pattern. This leads to the following constraints:

- When neuron i should be active in the pattern $\mu + 1$ i.e. $\xi_i^{\mu+1} = 0$, $\sum_j w_{ij}\xi_j^\mu > T + K$
- When neuron i should be inactive in the pattern $\mu + 1$ i.e. $\xi_i^{\mu+1} = 0$, $\sum_j w_{ij}\xi_j^\mu < T - K$

Whenever we have these constraints to not be satisfied, the modified perceptron rule dictates that we change the weights of active neurons only, by $+1$ or -1 in the neurons according to whether they should have been active in that corresponding pattern or not. Such a change of weights is continued for a certain number of iterations to allow the constraints to be satisfied. If not, the threshold and the weights are doubled and the learning continued, till the threshold reaches a limit. The maximal length of the sequence that can be stored in this manner is called the capacity of one perceptron. The capacity of the network is at that length of the sequence in the network, where we have at least half the neurons (perceptrons) able to learn the sequence that it is supposed to learn. At high number of neurons, there should be a sharp point where all of the neurons can learn all the patterns, and not a single neuron can learn, but due to low sizes in the simulations, we use this definition that is also mentioned in [1]. But for getting meaningful weight matrices, we need all the neurons in the network to learn the sequence that they are supposed to, which might be at a sequence length that is slightly less than the capacity. So it is at this sequence length that we look at the over-representation of bidirectional connectivity that we mentioned in 2.1. We also try out the same paradigm but the weights being learnt by finding a separatrix between the patterns to match their outputs, by reformulating the perceptron learning problem as a quadratic programming problem[4] and

calculate the statistics for the same. Theoretically, the quadratic programming problem should be able to converge to a solution whenever the perceptron learning algorithm can and vice versa.

3.2 Results

The capacity shown in figure 3.1show that at higher correlations between patterns, the capacity of the network increases. This finding is consistent with the finding of Clopath et al.[2] where we have an output set of patterns being learnt from an input set. The inputs and outputs are correlated among themselves. As an analogy to the situation that we have in learning correlated patterns in a sequence, we see that the input and output sets are from the same Markovian sequence, with the output set being circularly shifted. Hence, the input and output correlations which are mentioned in [2] are the same as the correlation we have between our patterns. At close

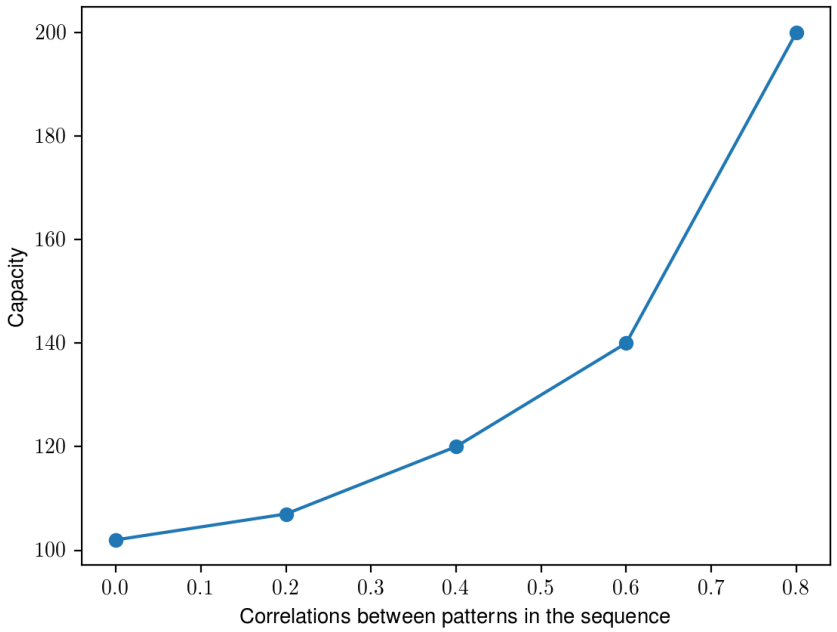

FIGURE 3.1: Capacity of the network storing a sequence, where weights are learnt using the perceptron learning rule, plotted with increasing correlations between patterns

to maximal capacity, we have that there is little overrepresentation of bidirectional connections, as shown in figure 3.2. We also tried plotting the same connectivity statistics in a network where the weights are learnt by formulating the problem as a quadratic programming problem. Here too, we observe little or no overrepresentation of bidirectional connections, as shown is figure 3.3.

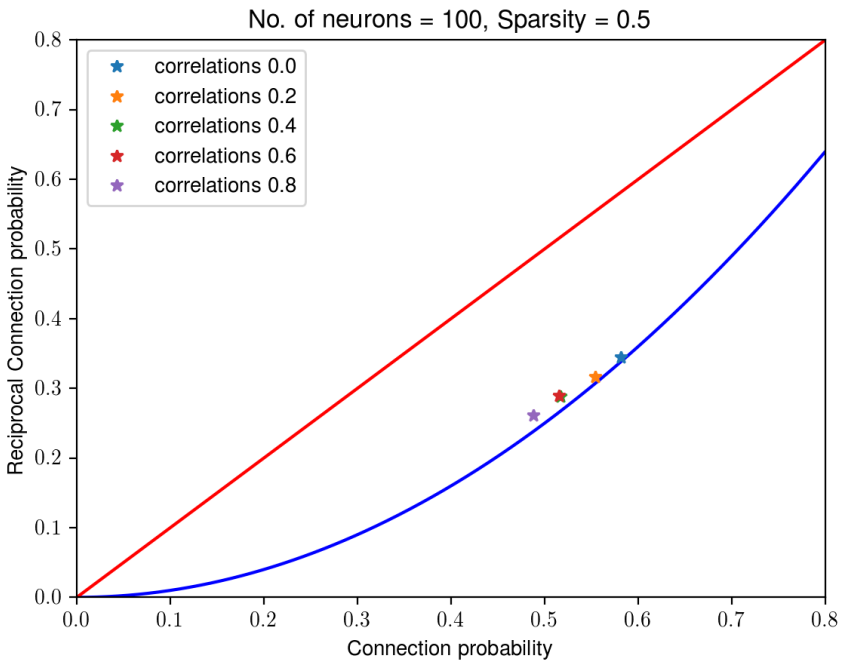

FIGURE 3.2: Bidrectional connectivity versus connection probability in a neural network storing a sequence, where weights are learnt by the perceptron learning rule

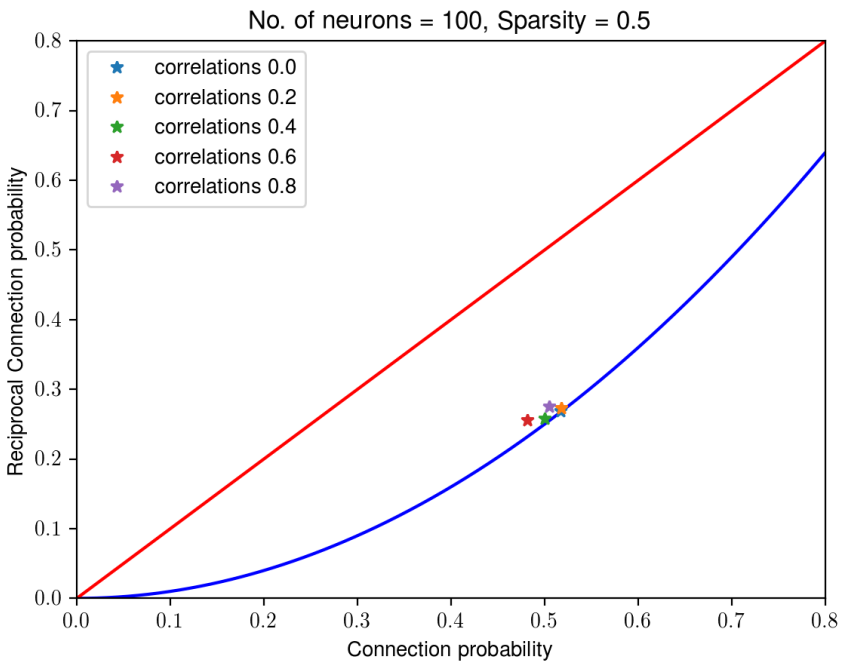

FIGURE 3.3: Bidrectional connectivity versus connection probability in a neural network storing a sequence, where weights are learnt by formulating and solving a quadratic learning problem

4 A neural network for learning a set of sequences

From the results on the learning of correlated sequences of patterns, we have seen that with increasing correlations, we should get eventually (though slowly as it varies with the square of the correlation) higher symmetricity in terms of the reciprocal weights between two neurons and to slowly increasing excess of bidirectionally connected neurons. Also, with changing the basin size, we see that we can move towards lower connection probabilities. However, due to possibly finite size effects - the fact that we have such a small length of the patterns, and quite low capacity for the appropriate basin size, we notice there is little change in the overrepresentation of bidirectionally connected neurons with increasing correlations. This led us to think that while it is correct that with increasing correlation between patterns, there should be an increasing correlation between reciprocal weights as explained in the previous chapter, in the limiting case, the sequence of highly correlated patterns would (assuming continuity) collapse to a single pattern, which would be like a single attractor in the attractor network. We have seen that in an attractor network, with low loads, there is significantly low correlation between reciprocal weights and the correlation between reciprocal weights saturates at some higher value as one approaches the capacity of the nwtwork. This is shown in the following simulation, in 4.1, 4.2 and 4.3. Taking the limiting case situation, we hypothesize that for optimality, like attractors

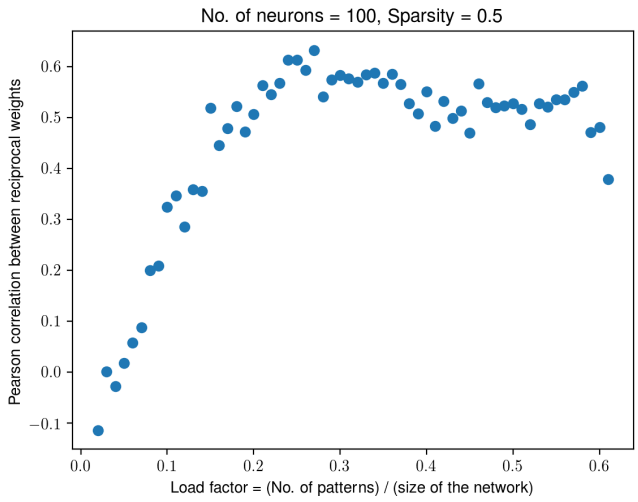

FIGURE 4.1: Increasing symmetricity with increasing load

at capacity, a *set* of correlated sequences (where each set consists of temporally correlated sequences, but between two sets, there are no correlations) might also be able to get close to the experimental data, in places where just highly correlated sequences cannot. The nature of information storage might be different from attractor networks or correlated sequences, but

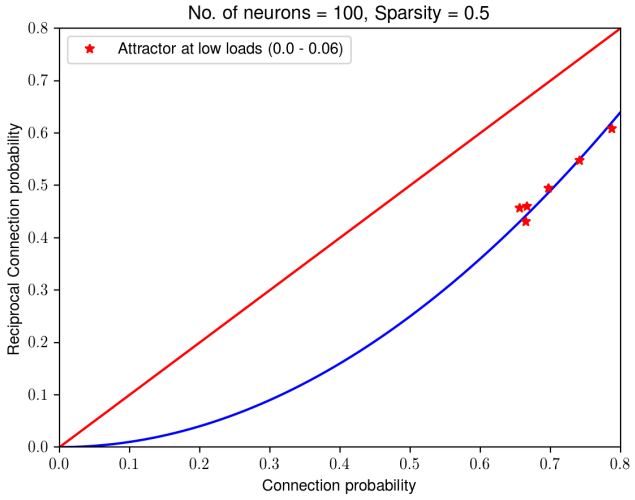

FIGURE 4.2: Increasing symmetricity with increasing load

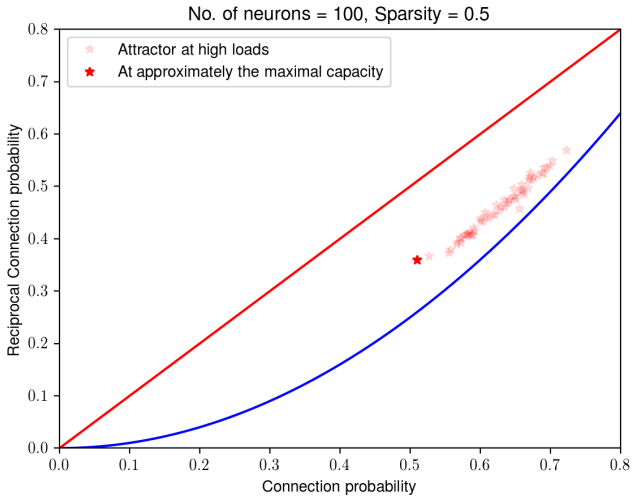

FIGURE 4.3: Increasing symmetricity with increasing load

rather something as a combination of both. A physical example that motivated the project initially can also be moulded to represent the set of correlated sequences paradigm. We had initially though of a correlated sequence as learning encodings of the arm as it moves in space, each encoding being correlated with the next, assuming that we go in discrete steps. If we think now of a set of sequences, we come to different arm sweeps, each sweep starting at a different uncorrelated time position but within a sweep, if we assume discrete motions, they are correlated with each other. Hence, we proceed to look at the situation where we have a set of correlated sequences.

4.1 Methods

The same methods of finding the capacity is used in this case, though we have not made a very thorough search of the parameter space. At capacity, the bidirectional connectivity statistics are shown. To get more realistic connection probabilities, we also changed the basin sizes and redid the same calculations.

4.2 Results

First, we look at different cases of a set of N sequences, with each sequence having L patterns. The results are shown in 4.4.

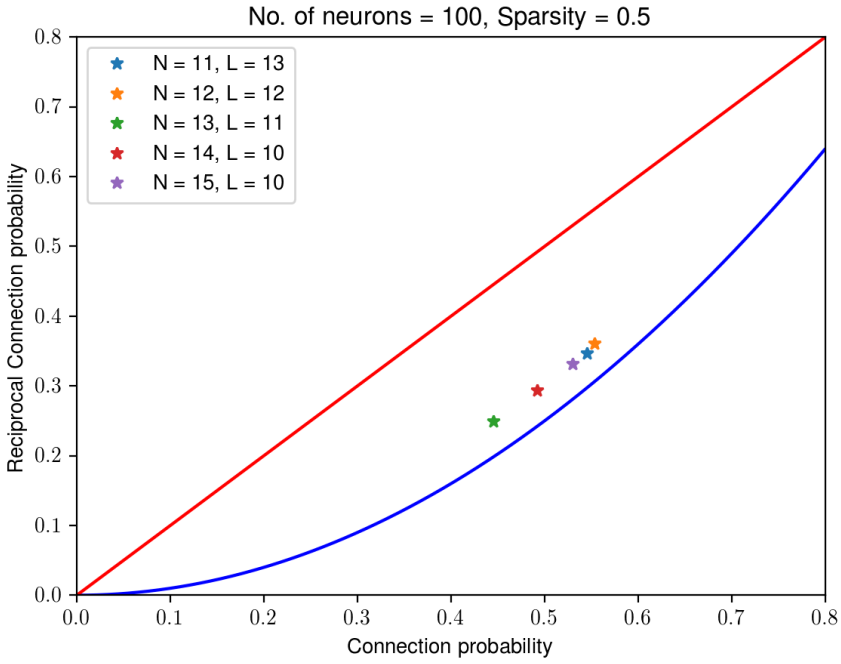

FIGURE 4.4: The calculation of reciprocal connection probability versus connection probability for the different values of N tried out

For the case of high basin sizes, we have the following result as shown is figure 4.5. We see that we can get closer to reproducing the experimental data.

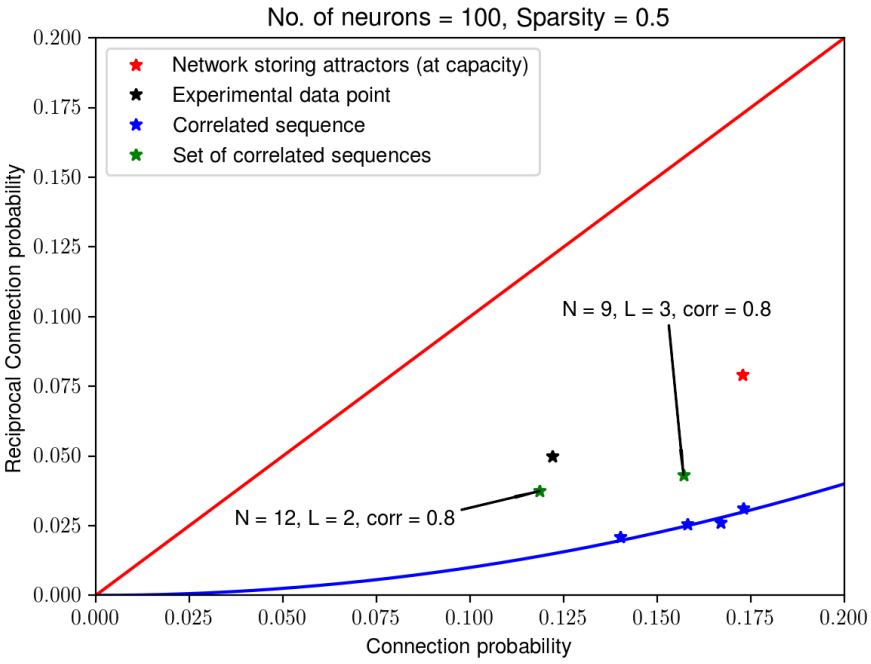

FIGURE 4.5: The calculation of reciprocal connection probability versus connection probability for the different values of N tried out, at high basin sizes

5 Discussions

We have provided here a different paradigm of what is meant by 'information' in terms of models of information storage in the neocortical circuitry. A lot of simulations and analytical studies need to be done with this paradigm to judge its efficacy. First and foremost, larger network sizes to remove the finite network effect should be simulated. With larger network sizes, capacities can be more precisely calculated and hence, we can judge the effectiveness of sets of correlated sequences in a better manner. At higher network sizes, it would also be easier to generate highly correlated sequences which are not exactly the same. Eventual tests of other parameters of the network needs to be done, like the in-degrees and out-degrees, as well as adding inhibition in a slightly different manner. The set of sequences paradigm deserves a closer look, especially in terms of searching the parameter space of the same. Finally, one might want to look at a spiking model implementation of the same to judge what might be going on in a situation closer to biology.

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